

COMPOSITE MATERIAL DATASHEETS CFRP T300/F593-PW

DEFENCE AND SPACE

Part: 1.4
Chapter: 1491
Revision: 1.2
Date: 24/01/2023
Page 1 of 3

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	T300/F593-PW
Commercial name	Hexcel Hexply W3T-282-42"-F593
Specifications	ABS5346 A (AIMS05-01-005 A)
Description	Carbon Fibre Reinforced Epoxy Prepreg Plain-Weave Fabric. Flow controlled, for sandwich applications.
Version	1.2 (24/01/2023)

REFERENCES

- TEA-T-NT-200081 Is. B "DERIVATION OF MATERIAL DESIGN VALUES FOR CFRP FABRIC PRE-PREG HEXCEL T300/F593"
- TEPA-NT-2310002 Is. A "A400M COWLINGS. CORRECTIONS IN DESIGN VALUES DUE TO ACCEPTABLE DEFECTS DURING MANUFACTURING. TEST RESULTS ANALYSIS."

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.237 mm
Density	1470 kg/m ³
Tg-onset (wet)	120 °C
Max. Operational Limit (MOL)	95 °C

Coefficients of Thermal Expansion (1/°C)	Range	-55..20°C	20..70°C	70..120°C
	α_1	4.3E-6	2.4E-06	2.4E-06
	α_2	4.3E-6	2.4E-06	2.4E-06
	α_3	5.9E-05	7.0E-05	8.8E-05

ELASTIC MODULI

Property	E ₁	E ₂	G ₁₂	V ₁₂	K _B
Cond. \ Units	MPa	MPa	MPa	-	-
T ≤ MOL (90°C)	50000	50000	2500	0.05	0.82
90°C < T ≤ 120°C			1000		

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E₁, E₂ and G₁₂ in buckling stability analyses

Property	E ₃	G ₁₃	G ₂₃	V ₁₃	V ₂₃
Cond. \ Units	MPa	MPa	MPa	-	-
T ≤ MOL (90°C)	9000	2600	2600	0.423	0.423
90°C < T ≤ 120°C	7800	1600	1600	0.157	0.157

PLAIN STRENGTHS

Property	Mean Values (MPa)				B-basis K _B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F _{1t}	558	558	558	-	0.85	1	1	-	474.3	474.3	474.3	-
F _{1c}	614	529	392	-	0.765	0.862	0.638	-	469.7	404.7	299.9	-
F _{2t}	558	558	558	-	0.85	1	1	-	474.3	474.3	474.3	-
F _{2c}	614	529	392	-	0.765	0.862	0.638	-	469.7	404.7	299.9	-
F ₁₂	125	90	77	60	0.831	0.72	0.616	0.48	103.9	74.8	64	49.9

Property	Mean Values (µε)				B-basis K _B	E-KDF			B-basis Values (µε)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
ε _{tall}	11200	11200	11200	11200	0.812	1	1	1	9100	9100	9100	9100
ε _{cail}	12300	10600	7800	5900	0.765	0.862	0.634	0.48	9400	8100	6000	4500

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{13} (MPa)	70	54	38	34	0.775	0.771	0.543	0.486	54.2	41.9	29.4	26.4
F_{3t} (MPa) (**)	70	50	-	-	0.83	0.71	-	-	58	41	-	-
G_{IC} (N/mm)	368	381	-	-	0.8	1.0	-	-	290	290	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
OHT_{QI}	198	198	198	-	0.833	1	1	-	165	165	165	-
OHC_{QI}	252	-	191	-	0.811	-	0.758	-	204.4	-	155	-
FHT_{QI}	196	196	196	-	0.842	1	1	-	165	165	165	-
FHC_{QI}	330	-	237	-	0.758	-	0.718	-	250	-	179.6	-
F_{BR-QI}	746	-	503	367	0.845	-	0.674	0.492	630	-	424.8	309.9
F_{PO} (*)	70	-	-	56.7	0.83	-	-	0.81	58	-	-	47

(*) Pull-out strength under tension load:

Allow. stress: $\tau_{PO} = EKDF \cdot K_B \cdot F_{PO} \cdot \Psi(t/d)$... with t thickness, d shank diameter, $\Psi = 1 + 0.6 \cdot (t/d - 1)$ if $t/d < 1$, otherwise $\Psi = 1$

Allow. force: $T_{PO} = \tau_{PO} \cdot \pi \cdot D_h \cdot t$... with t , again, the laminate thickness, and D_h the diameter of the bolt head

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
TAI_{QI}	4500	4500	4500	4500	0.833	1	1	1	3800	3800	3800	3800

$$E/t^2 = A \cdot (1 - e^{-B \cdot d_0/t})$$

Approximation model type: CDTM-T07

$$CAI = K_B \cdot EKDF \cdot CAI_{\infty-QI} \cdot (3 / K_{TOH})^C \cdot [1 + A / e^{B \cdot (E/t^2)}]$$

Property	Design Value
A (J/mm ²)	2.08
B	6.33

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ ($\mu\epsilon$)	3780	0.83	0.862	-	0.678
A	1.40				
B (mm ² /J)	0.61				
C	1.34				

...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_x/G_{xy} \}^{0.5}$$

(*) CAI shall be calculated not only for the 0° direction (X parallel to material 0° reference), but also for the rest of ply orientations, i.e. the 45° and 90° directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x , E_y , G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the allowable strains applicable to plies at 0° (CAI_0), at $\pm 45^\circ$ (CAI_{45}) and at 90° (CAI_{90}).

- Residual strength after transverse BVID near lightening holes

No design data for CAI_{hole} still available

- Residual strength after edge impact

No design data for CAI_E still available

SANDWICH WRINKLING

Parameters for wrinkling analysis of honeycomb sandwich panels with faces of this material (see SMM chapter 3510 "STRENGTH OF COMPOSITE SANDWICH PANELS").

Configuration	C ₁	C ₂	C ₃	C ₄
Default CMH-17 values (conservative)	0.247	0.078	0.33	0.0
Thin (2-3 plies) faces on Honeycomb Nomex - 32kg/m ³ - 4.8mm cell	0.422	0.0	0.569	0.0